Chapter 6. The Gamma and Zeta Functions

Analytic continuation and functional equations of Γ and ζ

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Overview

The gamma and zeta functions are among the most important non-elementary functions. The gamma function is ubiquitous; its basic structural property essentially characterizes it: $1/\Gamma(s)$ is the (simplest) entire function with zeros exactly at $s = 0, -1, -2, \ldots$. The zeta function is fundamental in analytic number theory, through Euler's identity

$$\prod_p \frac{1}{1-p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s)$$

over all primes. ζ is easily analytic for $\mathrm{Re}(s) > 1$; Riemann realized that the study of primes is bound up with its meromorphic continuation to $\mathbb{C}$. We prove this and the functional equation (a symmetry about $\mathrm{Re}(s) = 1/2$, based on the theta function identity), and estimate ζ near $\mathrm{Re}(s) = 1$ for the prime number theorem (Chapter 7). Following convention, the variable is s.

1 The gamma function

Motivation. The gamma function interpolates the factorial and is the main special function of analytic number theory. Its analytic continuation, its product formula and the identity $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ are the tools needed for the zeta function.

For $s > 0$ the **gamma function** is

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1}\, dt. \tag{1}$$

The integral converges: t^{s-1} is integrable near 0, and e^{-t} decays at infinity.

Proposition 1.1

The gamma function extends to an analytic function in the half-plane $\mathrm{Re}(s) > 0$, and is still given there by the integral formula (1).

Proof. It suffices to show the integral is holomorphic in each strip $S_{\delta,M} = \{\delta < \mathrm{Re}(s) < M\}$, $0 < \delta < M < \infty$. With $\sigma = \mathrm{Re}(s)$, $|e^{-t}t^{s-1}| = e^{-t}t^{\sigma-1}$, so $\Gamma(s) = \lim_{\epsilon\to 0} \int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\, dt$ converges for $s \in S_{\delta,M}$. Let $F_\epsilon(s) = \int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\, dt$, holomorphic in $S_{\delta,M}$ by Theorem 5.4. By Theorem 5.2 it suffices that $F_\epsilon \to \Gamma$ uniformly on the strip. Now

$$|\Gamma(s) - F_\epsilon(s)| \le \int_0^\epsilon e^{-t}t^{\sigma-1}\, dt + \int_{1/\epsilon}^\infty e^{-t}t^{\sigma-1}\, dt.$$

The first integral is $\le \epsilon^\delta/\delta$ for $0 < \epsilon < 1$, and the second is $\le \int_{1/\epsilon}^\infty e^{-t}t^{M-1}\, dt \le C\int_{1/\epsilon}^\infty e^{-t/2}\, dt \to 0$, uniformly in the strip. ∎

1.1 Analytic continuation

Although the integral does not converge absolutely elsewhere, there is a meromorphic function on $\mathbb{C}$ equal to Γ for $\mathrm{Re}(s) > 0$, its **analytic continuation**, still denoted Γ (unique, since the complement of the poles of a meromorphic function is connected).

Lemma 1.2

If $\operatorname{Re}(s) > 0$, then

$$\Gamma(s+1) = s\Gamma(s). \tag{2}$$

As a consequence $\Gamma(n+1) = n!$ for $n = 0, 1, 2, \ldots$.

Proof. Integrating by parts,

$$\int_\epsilon^{1/\epsilon} \frac{d}{dt}(e^{-t}t^s)\,dt = -\int_\epsilon^{1/\epsilon} e^{-t}t^s\,dt + s\int_\epsilon^{1/\epsilon} e^{-t}t^{s-1}\,dt,$$

and letting $\epsilon \to 0$ the left side vanishes, since $e^{-t}t^s \to 0$ as $t \to 0$ or $t \to \infty$. Also $\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1$; apply (2) successively. ∎

Theorem 1.3

The function $\Gamma(s)$ initially defined for $\operatorname{Re}(s) > 0$ has an analytic continuation to a meromorphic function on $\mathbb{C}$ whose only singularities are simple poles at the negative integers $s = 0, -1, \ldots$. The residue of Γ at $s = -n$ is $(-1)^n/n!$.

Proof. It suffices to extend Γ to each half-plane $\operatorname{Re}(s) > -m$, $m \geq 1$. For $\operatorname{Re}(s) > -m$ define

$$F_m(s) = \frac{\Gamma(s+m)}{(s+m-1)(s+m-2)\cdots s}.$$

F_m is meromorphic in $\operatorname{Re}(s) > -m$ with simple poles at $s = 0, -1, \ldots, -m+1$ and residues

$$\operatorname{res}_{s=-n} F_m(s) = \frac{\Gamma(-n+m)}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(m-n-1)!}{(m-1-n)!(-1)(-2)\cdots(-n)} = \frac{(-1)^n}{n!}.$$

By Lemma 1.2, $F_m = \Gamma$ for $\operatorname{Re}(s) > 0$, so by uniqueness $F_m = F_k$ $(1 \leq k \leq m)$ on the domain of F_k; this gives the continuation. ∎

Remark

By analytic continuation, $\Gamma(s+1) = s\Gamma(s)$ holds whenever $s \neq 0, -1, -2, \ldots$. At $s = -n$, $n \geq 1$, both sides are infinite and $\operatorname{res}_{s=-n}\Gamma(s+1) = -n\operatorname{res}_{s=-n}\Gamma(s)$. At $s = 0$, $\Gamma(1) = \lim_{s\to 0} s\Gamma(s)$.

An alternate proof. For $\operatorname{Re}(s) > 0$ split $\Gamma(s) = \int_0^1 e^{-t}t^{s-1}\,dt + \int_1^\infty e^{-t}t^{s-1}\,dt$. The second integral is entire; expanding e^{-t} and integrating termwise, $\int_0^1 e^{-t}t^{s-1}\,dt = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)}$, so

$$\Gamma(s) = \sum_{n=0}^\infty \frac{(-1)^n}{n!(n+s)} + \int_1^\infty e^{-t}t^{s-1}\,dt \qquad \text{for } \operatorname{Re}(s) > 0. \tag{3}$$

The series is meromorphic on $\mathbb{C}$ with poles at the negative integers and residue $(-1)^n/n!$ at $-n$: for fixed R, split it at $N > 2R$; the finite part is meromorphic in $|s| < R$ with the right poles and residues, and for $n > N$, $|s| < R$, $\left|\frac{(-1)^n}{n!(n+s)}\right| \leq \frac{1}{n!R}$, so the tail converges uniformly and is holomorphic there. Hence (3) holds on all of $\mathbb{C}$.

1.2 Further properties of Γ

Theorem 1.4

For all $s \in \mathbb{C}$,

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}. \tag{4}$$

$\Gamma(1-s)$ has simple poles at $s = 1, 2, \ldots$, so $\Gamma(s)\Gamma(1-s)$ is meromorphic with simple poles at all integers, as is $\pi/\sin \pi s$. By analytic continuation it suffices to prove (4) for $0 < s < 1$.

Lemma 1.5

For $0 < a < 1$, $\displaystyle\int_0^\infty \frac{v^{a-1}}{1+v}\, dv = \frac{\pi}{\sin \pi a}$.

Proof. With $v = e^x$, $\int_0^\infty \frac{v^{a-1}}{1+v}\, dv = \int_{-\infty}^\infty \frac{e^{ax}}{1+e^x}\, dx$, which equals $\pi/\sin \pi a$ by Example 2. ∎

Proof of Theorem 1.4. For $0 < s < 1$ and $t > 0$, the change of variables $u = vt$ gives $\Gamma(1-s) = \int_0^\infty e^{-u} u^{-s}\, du = t \int_0^\infty e^{-vt}(vt)^{-s}\, dv$. Hence

$$\begin{aligned}\Gamma(1-s)\Gamma(s) &= \int_0^\infty e^{-t} t^{s-1} \Gamma(1-s)\, dt = \int_0^\infty e^{-t} t^{s-1} \Big(t \int_0^\infty e^{-vt} (vt)^{-s}\, dv \Big) dt \\ &= \int_0^\infty \int_0^\infty e^{-t[1+v]} v^{-s}\, dt\, dv = \int_0^\infty \frac{v^{-s}}{1+v}\, dv = \frac{\pi}{\sin \pi(1-s)} = \frac{\pi}{\sin \pi s}.\end{aligned}$$

∎

With $s = 1/2$, since $\Gamma(s) > 0$ for $s > 0$, $\Gamma(1/2) = \sqrt{\pi}$.

Theorem 1.6

The function Γ has the following properties:
(i) $1/\Gamma(s)$ is an entire function of s with simple zeros at $s = 0, -1, -2, \ldots$ and it vanishes nowhere else.
(ii) $1/\Gamma(s)$ has growth $\left|\frac{1}{\Gamma(s)}\right| \le c_1 e^{c_2 |s| \log |s|}$. Therefore, $1/\Gamma$ is of order 1 in the sense that for every $\epsilon > 0$, there exists a bound $c(\epsilon)$ so that $\left|\frac{1}{\Gamma(s)}\right| \le c(\epsilon) e^{c_2 |s|^{1+\epsilon}}$.

Proof. By Theorem 1.4,

$$\frac{1}{\Gamma(s)} = \Gamma(1-s) \frac{\sin \pi s}{\pi}, \tag{5}$$

so the simple poles of $\Gamma(1-s)$ at $s = 1, 2, \ldots$ are cancelled by zeros of $\sin \pi s$, and $1/\Gamma$ is entire with simple zeros at $0, -1, -2, \ldots$.

For the estimate, first $\int_1^\infty e^{-t} t^\sigma\, dt \le e^{(\sigma+1)\log(\sigma+1)}$ for $\sigma > 0$: choose n with $\sigma \le n \le \sigma + 1$; then $\int_1^\infty e^{-t} t^\sigma\, dt \le \int_0^\infty e^{-t} t^n\, dt = n! \le n^n = e^{n \log n} \le e^{(\sigma+1)\log(\sigma+1)}$. Since (3) holds on $\mathbb{C}$, by (5)

$$\frac{1}{\Gamma(s)} = \Big(\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)} \Big) \frac{\sin \pi s}{\pi} + \Big(\int_1^\infty e^{-t} t^{-s}\, dt \Big) \frac{\sin \pi s}{\pi}.$$

$\left| \int_1^\infty e^{-t} t^{-s}\, dt \right| \le \int_1^\infty e^{-t} t^{|\sigma|}\, dt \le e^{(|s|+1)\log(|s|+1)}$ and $|\sin \pi s| \le e^{\pi |s|}$, so the second term is $\le c_1 e^{c_2 |s| \log |s|}$. For the first term: if $|\mathrm{Im}(s)| > 1$ it is $\le c e^{\pi |s|}$. If $|\mathrm{Im}(s)| \le 1$, let k be the integer with $k - 1/2 \le \mathrm{Re}(s) < k + 1/2$. If $k \ge 1$,

$$\sum_{n=0}^\infty \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi} = (-1)^{k-1} \frac{\sin \pi s}{\pi (k-1)!(k-s)} + \sum_{n \ne k-1} \frac{(-1)^n}{n!(n+1-s)} \frac{\sin \pi s}{\pi};$$

the first term is bounded since $\sin \pi s$ vanishes at $s = k$, and the second is majorized by $c\sum 1/n!$. If $k \le 0$ then $\mathrm{Re}(s) < 1/2$ and the series is again bounded by $c\sum 1/n!$. ∎

Theorem 1.7

For all $s \in \mathbb{C}$,

$$\frac{1}{\Gamma(s)} = e^{\gamma s} s \prod_{n=1}^{\infty} \Big(1 + \frac{s}{n}\Big) e^{-s/n}.$$

Here γ is **Euler's constant**, $\gamma = \lim_{N\to\infty} \big(\sum_{n=1}^{N} \frac{1}{n} - \log N\big)$. The limit exists: $\sum_{n=1}^{N} \frac{1}{n} - \log N = \sum_{n=1}^{N-1} \int_n^{n+1} \Big[\frac{1}{n} - \frac{1}{x}\Big] dx + \frac{1}{N}$, and by the mean value theorem $0 \le \frac{1}{n} - \frac{1}{x} \le \frac{1}{n^2}$ for $n \le x \le n+1$; so the sum is $\sum_{n=1}^{N-1} a_n + \frac{1}{N}$ with $0 \le a_n \le 1/n^2$, and $\sum a_n$ converges.

Proof. By Hadamard's theorem, since $1/\Gamma$ is entire of growth order 1 with simple zeros at $0, -1, -2, \dots$,

$$\frac{1}{\Gamma(s)} = e^{As+B} s \prod_{n=1}^{\infty} \Big(1 + \frac{s}{n}\Big) e^{-s/n}.$$

Since $s\Gamma(s) \to 1$ as $s \to 0$, $B = 0$ (or an integer multiple of $2\pi i$, which gives the same result). Putting $s = 1$ and using $\Gamma(1) = 1$,

$$e^{-A} = \prod_{n=1}^{\infty} \Big(1 + \frac{1}{n}\Big) e^{-1/n} = \lim_{N\to\infty} e^{\sum_{n=1}^{N} [\log(1+1/n) - 1/n]} = \lim_{N\to\infty} e^{-(\sum_{n=1}^{N} 1/n) + \log N + \log(1+1/N)} = e^{-\gamma}.$$

So $A = \gamma + 2\pi i k$; since $\Gamma(s)$ is real for real s, $k = 0$. ∎

Thus $1/\Gamma$ is essentially characterized (up to two normalizing constants) as the entire function with (i) simple zeros at $s = 0, -1, -2, \dots$ and no others, and (ii) order of growth 1. $\sin \pi s$ has a similar characterization (zeros at all integers), and satisfies the stricter $\sin \pi s = O(e^{c|s|})$; this estimate without the logarithm fails for $1/\Gamma$.

2 The zeta function

Motivation. The zeta function $\zeta(s) = \sum n^{-s}$ encodes the primes through the Euler product. To use complex analysis on it, we continue it to the whole plane (except a pole at $s = 1$) and prove the functional equation relating $\zeta(s)$ to $\zeta(1 - s)$.

The **Riemann zeta function** is initially defined for real $s > 1$ by $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$.

2.1 Functional equation and analytic continuation

Proposition 2.1

The series defining $\zeta(s)$ converges for $\mathrm{Re}(s) > 1$, and the function ζ is holomorphic in this half-plane.

Proof. For $s = \sigma + it$, $|n^{-s}| = |e^{-s \log n}| = e^{-\sigma \log n} = n^{-\sigma}$. If $\sigma \ge 1 + \delta > 1$ the series is uniformly bounded by $\sum 1/n^{1+\delta} < \infty$; so it converges uniformly on each half-plane $\mathrm{Re}(s) \ge 1 + \delta$ and defines a holomorphic function in $\mathrm{Re}(s) > 1$. ∎

The continuation is subtler than for Γ; it relates ζ to the **theta function** $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$, $t > 0$, which satisfies (by Poisson summation, (4) of Chapter 4) $\vartheta(t) = t^{-1/2}\vartheta(1/t)$. We need

$$\vartheta(t) \le C t^{-1/2} \quad \text{as } t \to 0, \qquad |\vartheta(t) - 1| \le C e^{-\pi t} \quad \text{for } t \ge 1 :$$

the first follows from the functional equation, the second from $\sum_{n\ge1} e^{-\pi n^2 t} \le \sum_{n\ge1} e^{-\pi n t} \le Ce^{-\pi t}$ for $t \ge 1$.

Theorem 2.2

If $\mathrm{Re}(s) > 1$, then

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^\infty u^{(s/2)-1}[\vartheta(u) - 1]\,du.$$

Proof. The key is

$$\int_0^\infty e^{-\pi n^2 u} u^{(s/2)-1}\,du = \pi^{-s/2}\Gamma(s/2)n^{-s}, \qquad n \ge 1 : \tag{6}$$

with $u = t/\pi n^2$ the left side becomes $\left(\int_0^\infty e^{-t}t^{(s/2)-1}\,dt\right)(\pi n^2)^{-s/2}$. Since $\frac{\vartheta(u)-1}{2} = \sum_{n=1}^\infty e^{-\pi n^2 u}$, and the estimates on ϑ justify interchanging sum and integral,

$$\frac{1}{2}\int_0^\infty u^{(s/2)-1}[\vartheta(u) - 1]\,du = \sum_{n=1}^\infty \int_0^\infty u^{(s/2)-1}e^{-\pi n^2 u}\,du = \pi^{-s/2}\Gamma(s/2)\sum_{n=1}^\infty n^{-s} = \pi^{-s/2}\Gamma(s/2)\zeta(s).$$

■

The **xi function**, a more symmetric modification of ζ, is defined for $\mathrm{Re}(s) > 1$ by

$$\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s). \tag{7}$$

Theorem 2.3

The function ξ is holomorphic for $\mathrm{Re}(s) > 1$ and has an analytic continuation to all of $\mathbb{C}$ as a meromorphic function with simple poles at $s = 0$ and $s = 1$. Moreover, $\xi(s) = \xi(1 - s)$ for all $s \in \mathbb{C}$.

Proof. Heuristically: multiply $\sum_n e^{-\pi n^2 u} = u^{-1/2}\sum_n e^{-\pi n^2/u}$ by $u^{(s/2)-1}$ and integrate, using (6) and the change $u \mapsto 1/u$ (discarding the $n = 0$ terms). Precisely: let $\psi(u) = [\vartheta(u) - 1]/2$. The functional equation of ϑ gives

$$\psi(u) = u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}.$$

By Theorem 2.2, for $\mathrm{Re}(s) > 1$,

$$\begin{aligned}
\pi^{-s/2}\Gamma(s/2)\zeta(s) &= \int_0^1 u^{(s/2)-1}\psi(u)\,du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \int_0^1 u^{(s/2)-1}\Big[u^{-1/2}\psi(1/u) + \frac{1}{2u^{1/2}} - \frac{1}{2}\Big]du + \int_1^\infty u^{(s/2)-1}\psi(u)\,du \\
&= \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du
\end{aligned}$$

(substituting $u \mapsto 1/u$ in the first integral). Therefore

$$\xi(s) = \frac{1}{s-1} - \frac{1}{s} + \int_1^\infty (u^{(-s/2)-1/2} + u^{(s/2)-1})\psi(u)\,du.$$

Since ψ decays exponentially at infinity, the integral is entire, so ξ continues to $\mathbb{C}$ with simple poles at $s = 0$, $s = 1$. The integral is unchanged under $s \mapsto 1 - s$, and so is $\frac{1}{s-1} - \frac{1}{s}$; hence $\xi(s) = \xi(1-s)$.
■

Theorem 2.4

The zeta function has a meromorphic continuation into the entire complex plane, whose only singularity is a simple pole at $s = 1$.

Proof. By (7), $\zeta(s) = \pi^{s/2}\xi(s)/\Gamma(s/2)$. $1/\Gamma(s/2)$ is entire with simple zeros at $0, -2, -4, \ldots$, so the simple pole of ξ at 0 is cancelled, and the only singularity of ζ is a simple pole at $s = 1$. ■

A more elementary approach gives the continuation to $\mathrm{Re}(s) > 0$ and is useful for the growth of ζ near $\mathrm{Re}(s) = 1$: compare $\sum n^{-s}$ with $\int_1^\infty x^{-s}\,dx$.

Proposition 2.5

There is a sequence of entire functions $\{\delta_n(s)\}_{n=1}^\infty$ that satisfy the estimate $|\delta_n(s)| \le |s|/n^{\sigma+1}$, where $s = \sigma + it$, and such that

$$\sum_{1\le n<N} \frac{1}{n^s} - \int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \delta_n(s), \tag{8}$$

whenever N is an integer > 1.

Proof. Set

$$\delta_n(s) = \int_n^{n+1} \Big[\frac{1}{n^s} - \frac{1}{x^s}\Big]dx. \tag{9}$$

By the mean-value theorem applied to x^{-s}, $|\frac{1}{n^s} - \frac{1}{x^s}| \le \frac{|s|}{n^{\sigma+1}}$ for $n \le x \le n+1$, so $|\delta_n(s)| \le |s|/n^{\sigma+1}$; and $\int_1^N \frac{dx}{x^s} = \sum_{1\le n<N} \int_n^{n+1} \frac{dx}{x^s}$. ■

Corollary 2.6

For $\mathrm{Re}(s) > 0$ we have

$$\zeta(s) - \frac{1}{s-1} = H(s),$$

where $H(s) = \sum_{n=1}^\infty \delta_n(s)$ is holomorphic in the half-plane $\mathrm{Re}(s) > 0$.

Proof. First let $\mathrm{Re}(s) > 1$ and $N \to \infty$ in (8): $\int_1^N x^{-s}dx \to \frac{1}{s-1}$, and by $|\delta_n(s)| \le |s|/n^{\sigma+1}$ the series $\sum \delta_n(s)$ converges uniformly on compact subsets of $\mathrm{Re}(s) \ge \delta > 0$. This proves the identity for $\mathrm{Re}(s) > 1$; the uniform convergence shows H is holomorphic in $\mathrm{Re}(s) > 0$, so ζ extends there and the identity persists. ■

For $\mathrm{Re}(s) > 1$, $|\zeta(s)| \le \sum n^{-\sigma}$, so ζ is bounded in $\mathrm{Re}(s) \ge 1 + \delta$. The growth near $\mathrm{Re}(s) = 1$ is mild (the following estimates are crude but suffice for Chapter 7).

Proposition 2.7

Suppose $s = \sigma + it$ with $\sigma, t \in \mathbb{R}$. Then for each σ_0, $0 \le \sigma_0 \le 1$, and every $\epsilon > 0$, there exists a constant c_ϵ so that (i) $|\zeta(s)| \le c_\epsilon |t|^{1-\sigma_0+\epsilon}$, if $\sigma_0 \le \sigma$ and $|t| \ge 1$.
(ii) $|\zeta'(s)| \le c_\epsilon |t|^\epsilon$, if $1 \le \sigma$ and $|t| \ge 1$.

In particular $\zeta(1 + it) = O(|t|^\epsilon)$ as $|t| \to \infty$, and likewise for ζ'.

Proof. Besides $|\delta_n(s)| \le |s|/n^{\sigma+1}$ we have $|\delta_n(s)| \le 2/n^\sigma$, from (9) and $|n^{-s}| = n^{-\sigma}$, $|x^{-s}| \le n^{-\sigma}$ for $x \ge n$. Combining via $A = A^\delta A^{1-\delta}$ $(0 \le \delta \le 1)$,

$$|\delta_n(s)| \le \Big(\frac{|s|}{n^{\sigma_0+1}}\Big)^\delta \Big(\frac{2}{n^{\sigma_0}}\Big)^{1-\delta} \le \frac{2|s|^\delta}{n^{\sigma_0+\delta}} \qquad (\sigma \ge \sigma_0).$$

Take $\delta = 1 - \sigma_0 + \epsilon$ and use Corollary 2.6: for $\sigma \geq \sigma_0$,

$$|\zeta(s)| \leq \Big|\frac{1}{s-1}\Big| + 2|s|^{1-\sigma_0+\epsilon} \sum_{n=1}^{\infty} \frac{1}{n^{1+\epsilon}},$$

which gives (i). For (ii), by the Cauchy integral formula on the circle of radius r about s,

$$\zeta'(s) = \frac{1}{2\pi r} \int_0^{2\pi} \zeta(s + re^{i\theta}) e^{-i\theta}\, d\theta.$$

Take $r = \epsilon$; this circle lies in $\mathrm{Re} \geq 1 - \epsilon$, and (ii) follows from (i) (with $\sigma_0 = 1 - \epsilon$ and 2ϵ in place of ϵ). ■